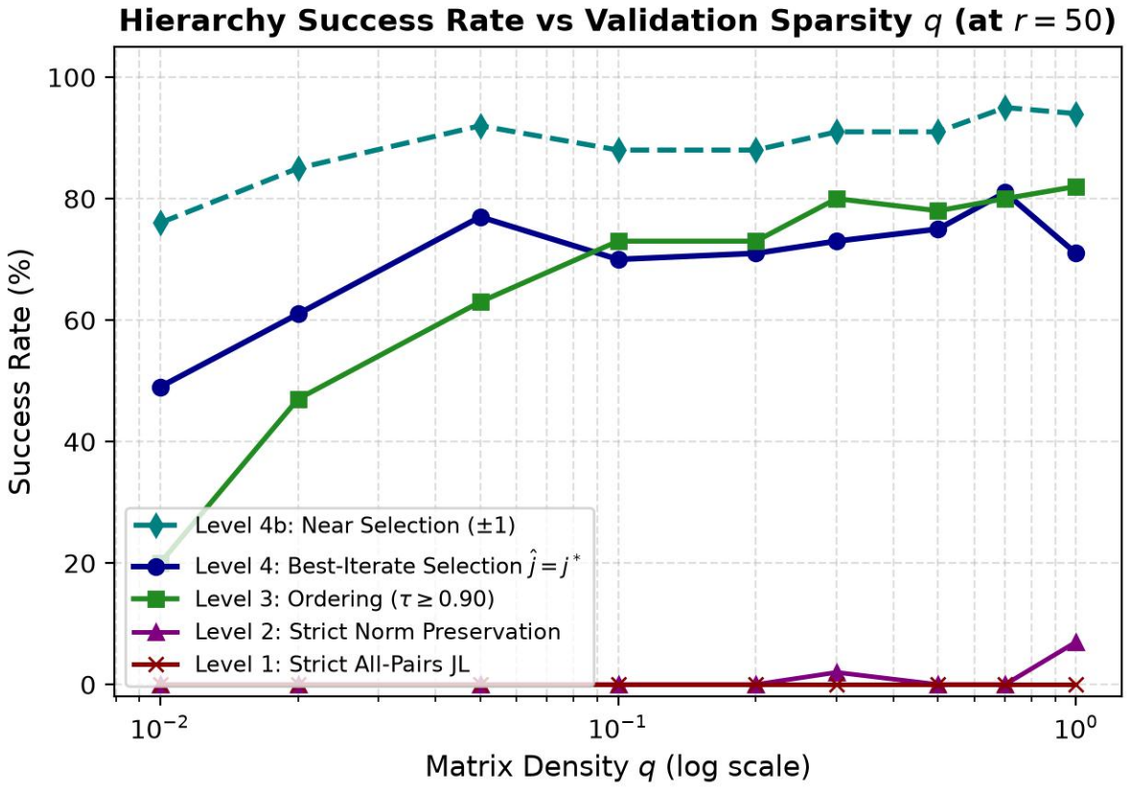
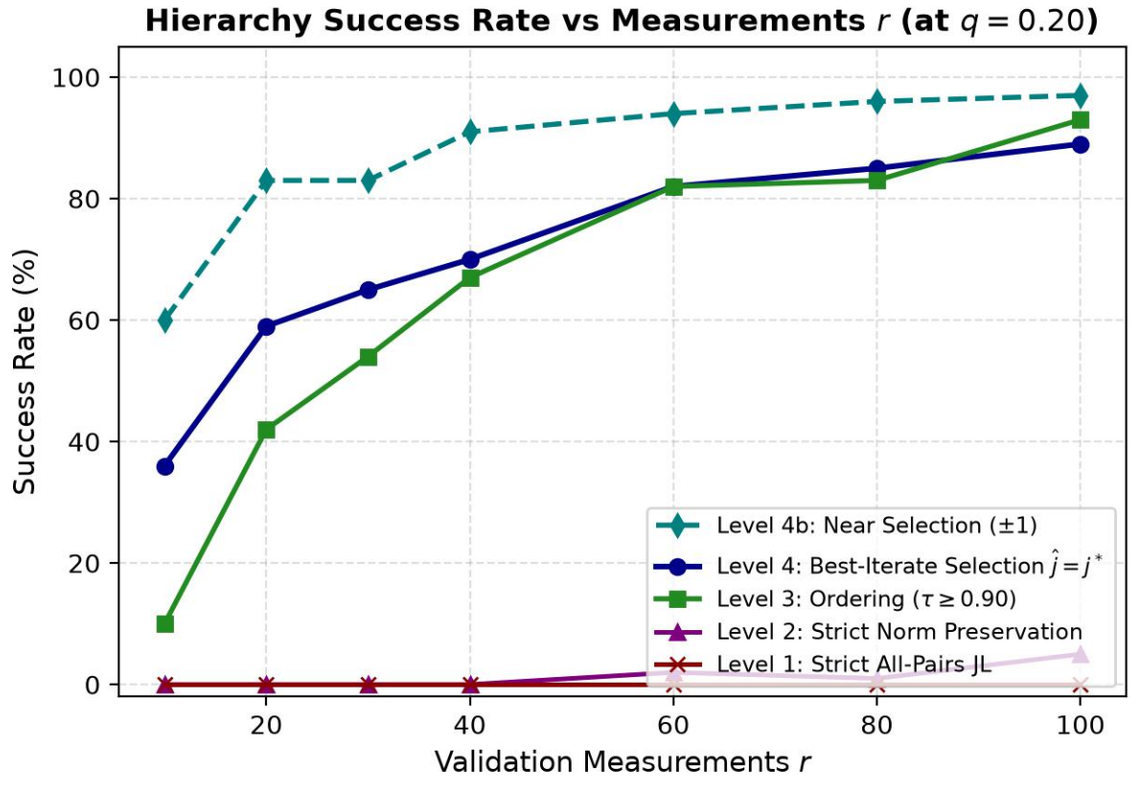
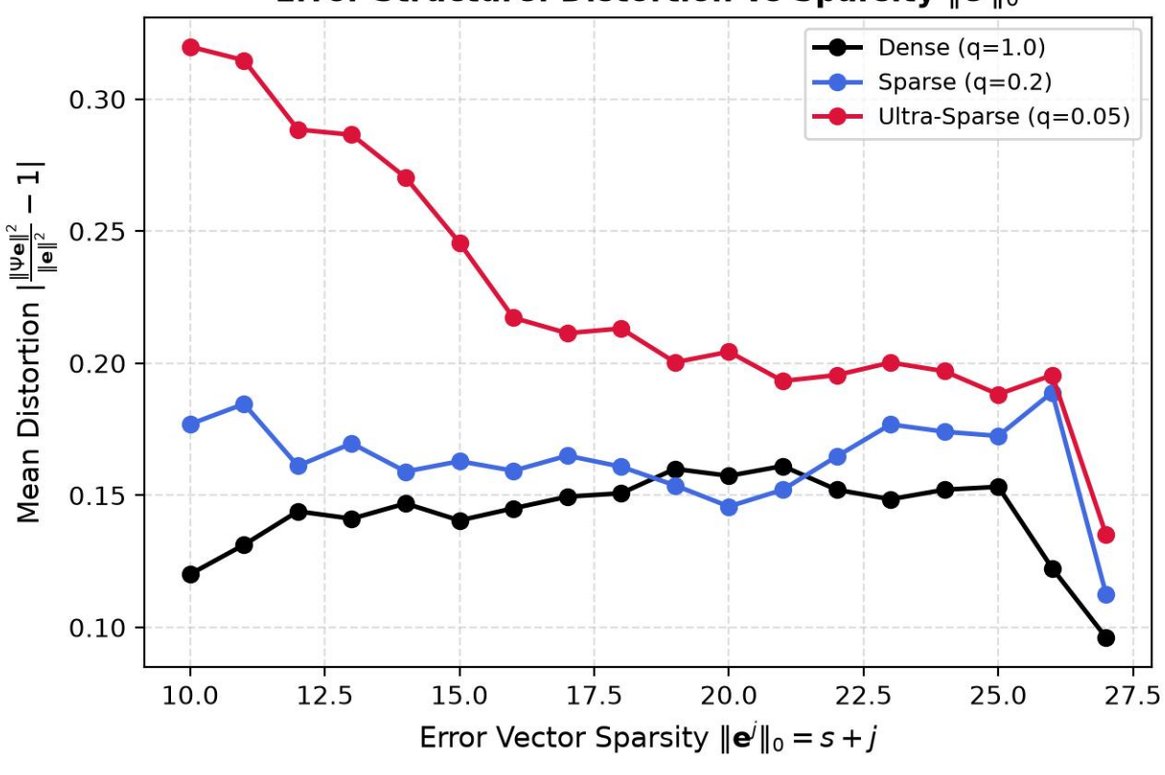
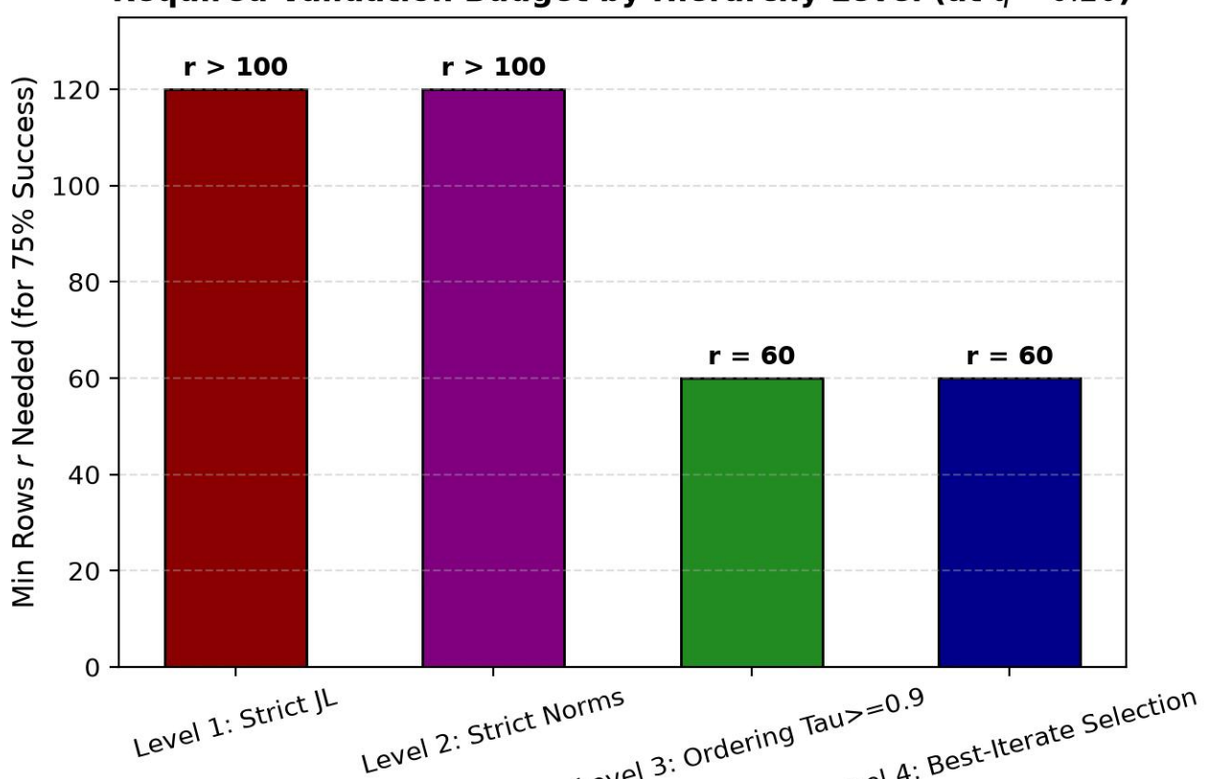

Hierarchy Success Rate vs Validation Sparsity q (at $r = 50$)
Success Rate (%)
Matrix Density q (log scale)
Level 4b: Near Selection (±1)
Level 4: Best-Iterate Selection $\hat{j} = j^*$
Level 3: Ordering ($\tau \geq 0.90$)
Level 2: Strict Norm Preservation
Level 1: Strict All-Pairs JL


Hierarchy Success Rate vs Measurements r (at $q = 0.20$)
Success Rate (%)
Validation Measurements r
Level 4b: Near Selection (±1)
Level 4: Best-Iterate Selection $\hat{j} = j^*$
Level 3: Ordering ($\tau \geq 0.90$)
Level 2: Strict Norm Preservation
Level 1: Strict All-Pairs JL


Error Structure: Distortion vs Sparsity $\|e^j\|_0$
Mean Distortion $\left|\frac{\|\Psi e\|^2}{\|e\|^2} - 1\right|$
Error Vector Sparsity $\|e^j\|_0 = s + j$
Dense (q=1.0)
Sparse (q=0.2)
Ultra-Sparse (q=0.05)


Required Validation Budget by Hierarchy Level (at $q = 0.20$)
Min Rows r Needed (for 75% Success)
r > 100
r > 100
r = 60
r = 60
Level 1: Strict JL
Level 2: Strict Norms
Level 3: Ordering Tau>=0.9
Level 4: Best-Iterate Selection